

Automated Sleep Staging

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11/14/2025



Agenda

- 1. Challenge presentation**
- 2. Primer on Sleep EEG and and sleep staging**
- 3. Technical elements on the challenge**

Powering precision medicine for the brain

Beacon's EEG neurobiomarker platform is engineered to accelerate clinical trials and enable new treatments for patients with neurological and psychiatric disease.

Biomarker Applications



Treatment Efficacy

Quantitative assessment of treatment impact on EEG and sleep data ensure clinical endpoints are reached



PK/PD & Dose Selection

Understand PK/PD relationships with brain activity to identify doses with optimal efficacy and safety profiles



Population Stratification

Identify optimal patient populations to understand and select cohorts

ENABLED BY OUR END-TO-END AUGMENTED EEG PLATFORM



DREEM HEADBAND



STATE OF THE ART
MACHINE LEARNING



WORLD-CLASS CLINICO-
EEG DATABASE

Dreem Headband: PSG quality in a lightweight, easy to use device



EEG sensors

5 EEG sensors to measure EEG:

- A.** 2 frontal sensors in F7 and F8 locations to measure frontal brain activity
- B.** 1 ground sensor on the frontal band (Fp2 location)
- C.** 2 occipital sensors on O1 and O2 locations to monitor occipital brain activity



Audio

- D.** Bone conduction speaker for audio output



Accelerometer

- E.** 3D accelerometer to measure movements, head position, and respiratory rate/trace during sleep



Miscellaneous

- F.** Power Button
- G.** Magnet port for charging

Challenge: Automated sleep staging on PSG Data

Data from 12 records, 8 for training and 4 for evaluation.

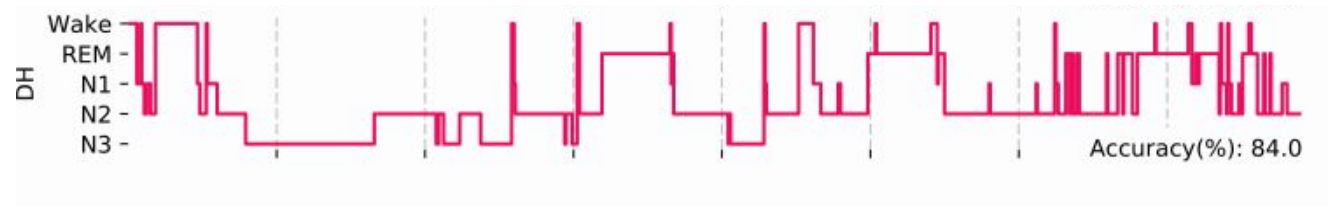
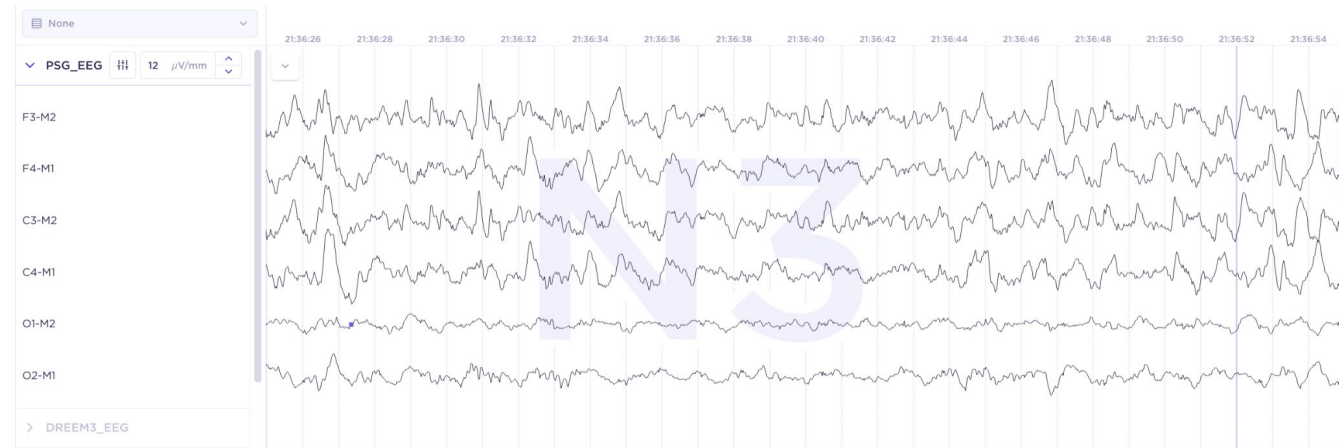
Each record contains data from a complete night and from the following biosignals:

- 5 EEG channels (sampled at 100 Hz)

Goal: Predict the sleep stage of each section of 30 seconds (epoch) of data.

Target: Sleep stage corresponding to each section of 30 seconds of signal.

Sleep stages: Wake (0), N1 (1), N2 (2), N3 (3), REM (4), NS (-1)



Why sleep staging matter

Nocturnal Sleep Dynamics Identify Narcolepsy Type 1

[Fabio Pizza](#)^{1,2}, [Stefano Vandi](#)^{1,2}, [Martina Iloti](#)¹, [Christian Franceschini](#)³, [Rocco Liguori](#)^{1,2}, [Emmanuel Mignot](#)^{1,2,∞}, [Giuseppe Plazzi](#)^{1,2,∞}

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Article | [Open access](#) | Published: 29 February 2024

Sleep disorders in Parkinson’s disease, an early and multiple problem

[Pauline Dodet](#) ✉, [Marion Houot](#), [Smaranda Leu-Semenescu](#), [Jean-Christophe Corvol](#), [Stéphane Lehericy](#), [Graziella Mangone](#), [Marie Vidailhet](#), [Emmanuel Roze](#) & [Isabelle Arnulf](#)

[npj Parkinson's Disease](#) **10**, Article number: 46 (2024) | [Cite this article](#)

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Alzheimer’s Disease phenotypes show different sleep architecture

[Neus Falgàs](#)^{1,2,3,*}, [Christine M Walsh](#)^{1,*}, [Leslie Yack](#)^{1,4}, [Alexander J Simon](#)¹, [Isabel E Allen](#)^{1,2}, [Joel H Kramer](#)^{1,6}, [Howard J Rosen](#)^{1,2}, [Renaud La Joie](#)¹, [Gil Rabinovici](#)¹, [Bruce Miller](#)^{1,2}, [Salvatore Spina](#)^{1,2}, [William W Seeley](#)¹, [Kamalini Ranasinghe](#)¹, [Keith Vossel](#)⁵, [Thomas C Neylan](#)^{1,6,†}, [Lea T Grinberg](#)^{1,2,7,†}

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PMCID: PMC10404632 NIHMSID: NIHMS1864777 PMID: [36749893](#)

Article | [Open access](#) | Published: 06 December 2018

Neural network analysis of sleep stages enables efficient diagnosis of narcolepsy

[Jens B. Stephansen](#), [Alexander N. Olesen](#), [Mads Olsen](#), [Aditya Ambati](#), [Eileen B. Leary](#), [Hyatt E. Moore](#), [Oscar Carrillo](#), [Ling Lin](#), [Fang Han](#), [Han Yan](#), [Yun L. Sun](#), [Yves Dauvilliers](#), [Sabine Scholz](#), [Lucie Barateau](#), [Birgit Hogg](#), [Ambra Stefani](#), [Seung Chul Hong](#), [Tae Won Kim](#), [Fabio Pizza](#), [Giuseppe Plazzi](#), [Stefano Vandi](#), [Elena Antelmi](#), [Dimitri Perrin](#), [Samuel T. Kuna](#), ... [Emmanuel Mignot](#) ✉ [+ Show authors](#)

[Nature Communications](#) **9**, Article number: 5229 (2018) | [Cite this article](#)

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Article | [Open access](#) | Published: 26 September 2025

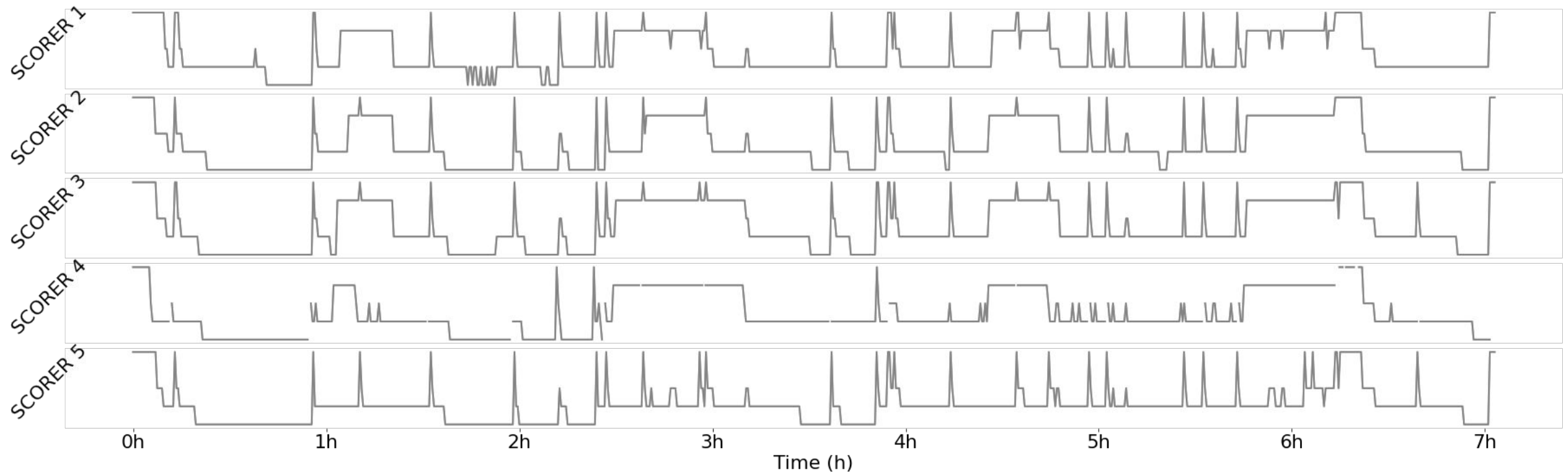
Sleep stage mixing is associated with poor prognosis in early Parkinson’s disease

[Pauline Dodet](#) ✉, [Emmanuel During](#), [Isabelle Arnulf](#), [Claudia Trenkwalder](#), [Brit Mollenhauer](#), [Friederike Sixel-Döring](#), [Emmanuel Roze](#), [Marie Vidailhet](#), [Thomas Andrillon](#), [Stéphane Lehericy](#), [Jean-Christophe Corvol](#), [Graziella Mangone](#), [Smaranda Leu-Semenescu](#), [Emmanuel Mignot](#), [Poul Jennum](#) & [Andreas Brink-Kjaer](#) ✉

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Limitations of existing approaches



Scorers from different sleep centers and background will often score sleep slightly differently, inter-rater agreement is around 82% and lower on more pathological populations.

Primer on sleep staging (AASM guidelines)

Wake:

Alpha rhythm (posterior dominant rhythm) over the occipital region (individuals generating alpha rhythm with eye closure) b. Other findings consistent with stage W (all individuals)

- i. Eye blinks (0.5 to 2 Hz)
- ii. Rapid eye movements associated with normal or high chin muscle tone i
- ii. Reading eye movements

NREM1

In individuals who do not generate alpha rhythm, score stage N1 commencing with the earliest of ANY of the following phenomena: a. EEG activity in range of 4-7 Hz with slowing of background frequencies by ≥ 1 Hz from those of stage W b. Vertex sharp waves c. Slow eye movements

NREM2

- a. One or more K complexes unassociated with arousals (A well-delineated, negative, sharp wave immediately followed by a positive component standing out from the background EEG)
- b. One or more sleep spindles (A train of distinct sinusoidal waves with frequency 11-16 Hz (most commonly 12-14 Hz))

NREM3

Slow wave activity: Waves of frequency 0.5 Hz-2 Hz and peak-to-peak amplitude >75 μ V, measured over the frontal regions referenced to the contralateral ear or mastoid (F4-M1, F3-M2).

REM:

Low-amplitude, mixed-frequency (LAMF) EEG activity without K complexes or sleep spindles b. Low chin EMG tone for the majority of the epoch and concurrent with REMs c. REMs at any position within the epoch

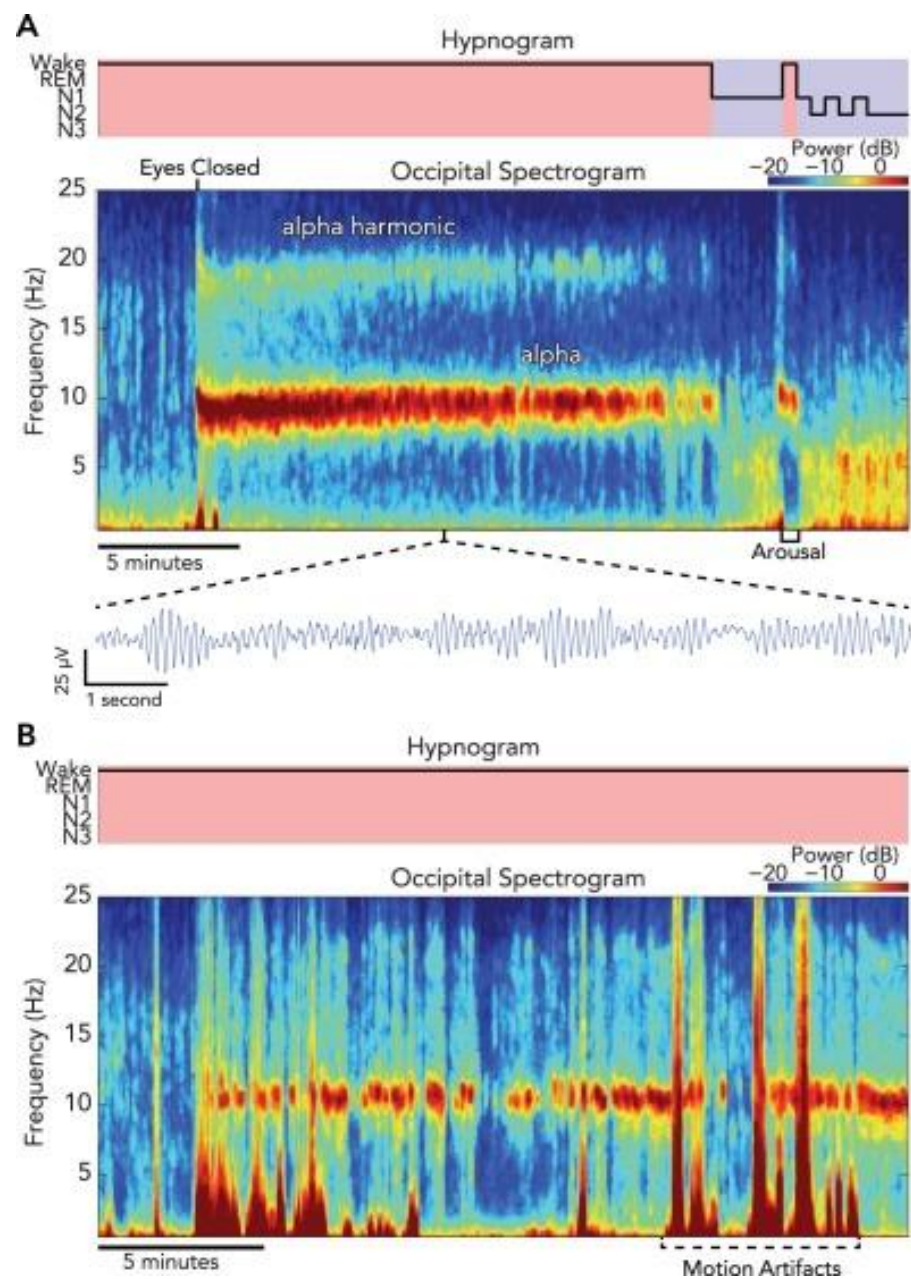
Sleep Neurophysiological Dynamics Through the Lens of Multitaper Spectral Analysis

Michael J Prerau¹, Ritchie E Brown², Matt T Bianchi³, Jeffrey M Ellenbogen⁴,
Patrick L Purdon¹

Affiliations + expand

PMID: 27927806 PMCID: [PMC5343535](#) DOI: [10.1152/physiol.00062.2015](#)

WAKE



WAKE

Eyes Open (Quiescent)

Delta	Low power
Theta	Low power
Alpha	Low power
Sigma	Low power
Background	Low power

Eyes Closed

Delta	Low power, below REM levels Low frequency noise possible
Theta	Low power, below REM levels
Alpha	High power, clear oscillation, possible harmonic at double the alpha frequency
Sigma	Low power
Background	Low power

Eyes Open (Active)

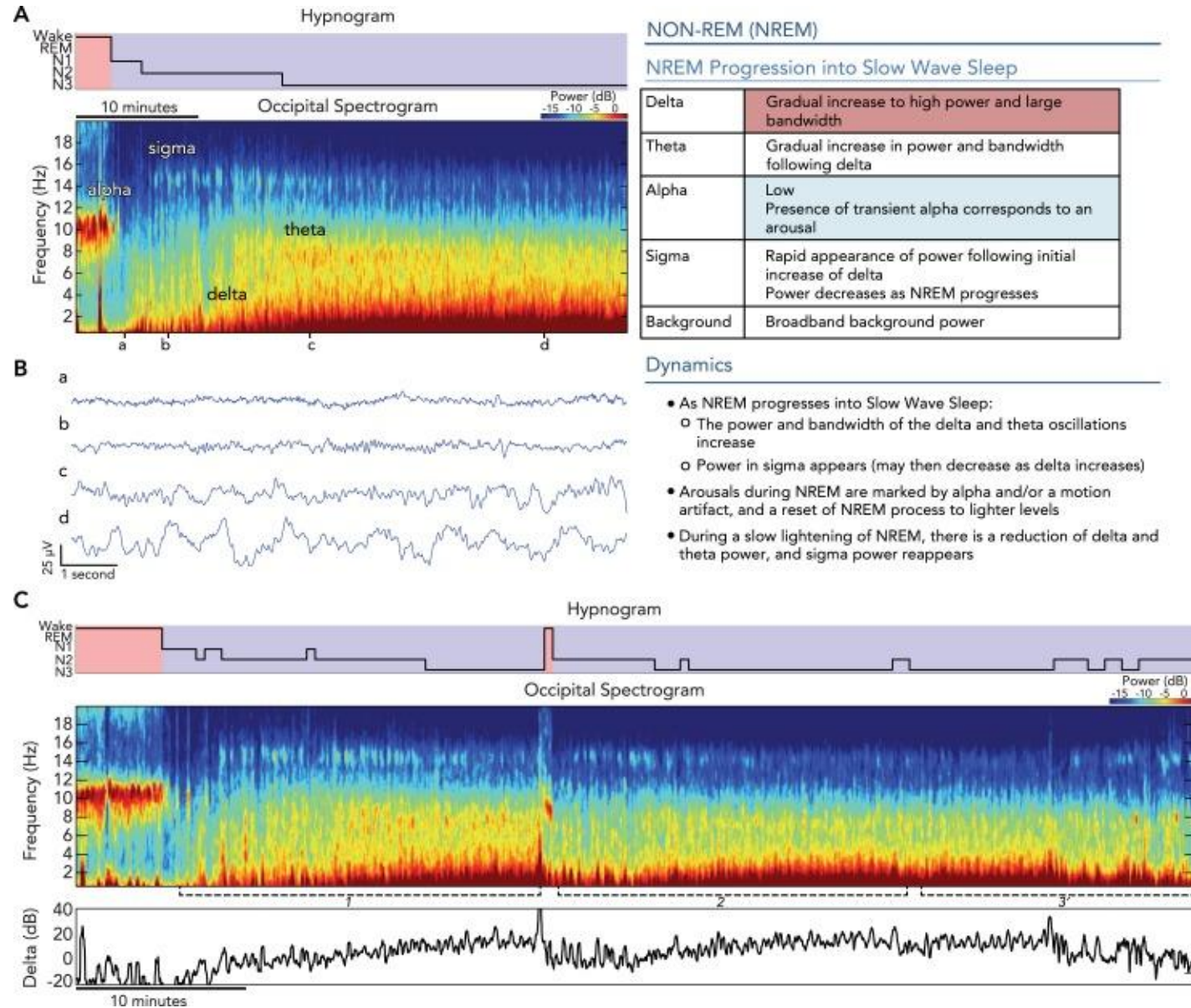
Delta	Low power
Theta	Low power
Alpha	Sporadic bursts or absent
Sigma	Low power
Background	Broadband background power, motion artifacts

Dynamics

- Eyes open wake has low power at all frequencies and no clear oscillatory dynamics
- Eyes closed wake shows a strong alpha oscillation which decreases in power and bandwidth during the sleep onset process
- Active wake has a mix of eyes closed, eyes open, and motion artifact
- Motion artifacts are seen as distinct vertical high power bands and may be seen during any stage
- Arousals are seen as a transient burst of power in alpha and may be seen during any stage



NREM



Spindles in NREM

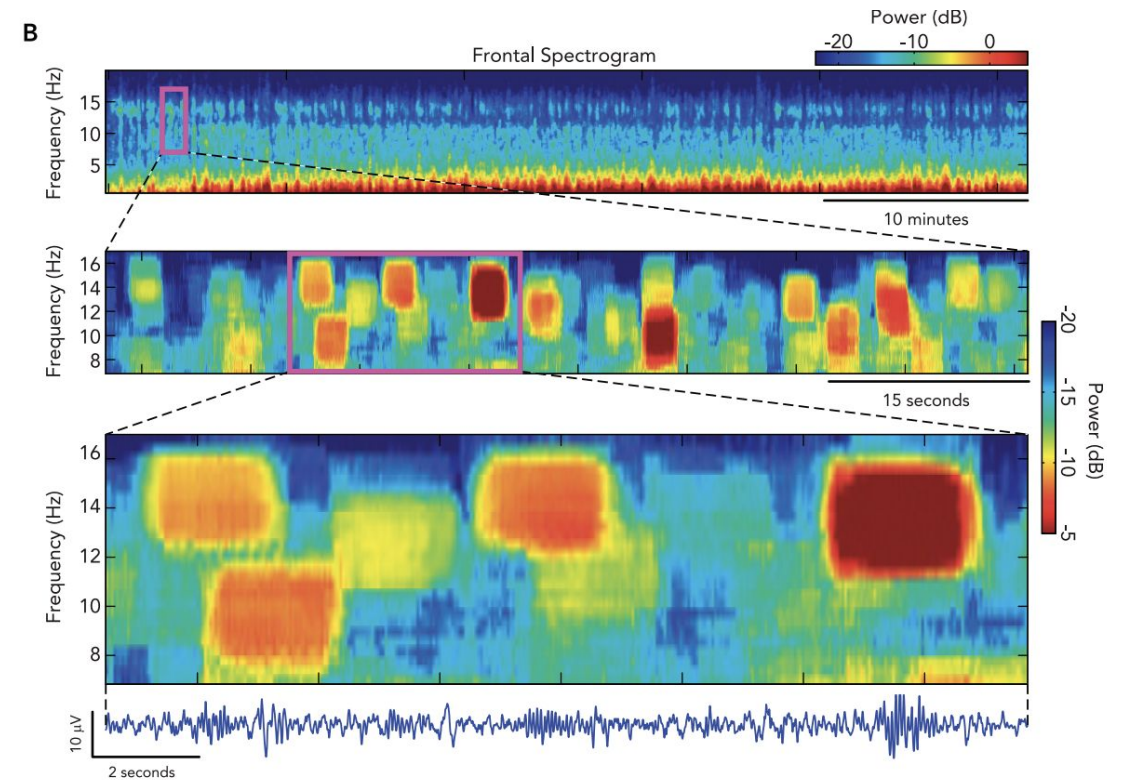
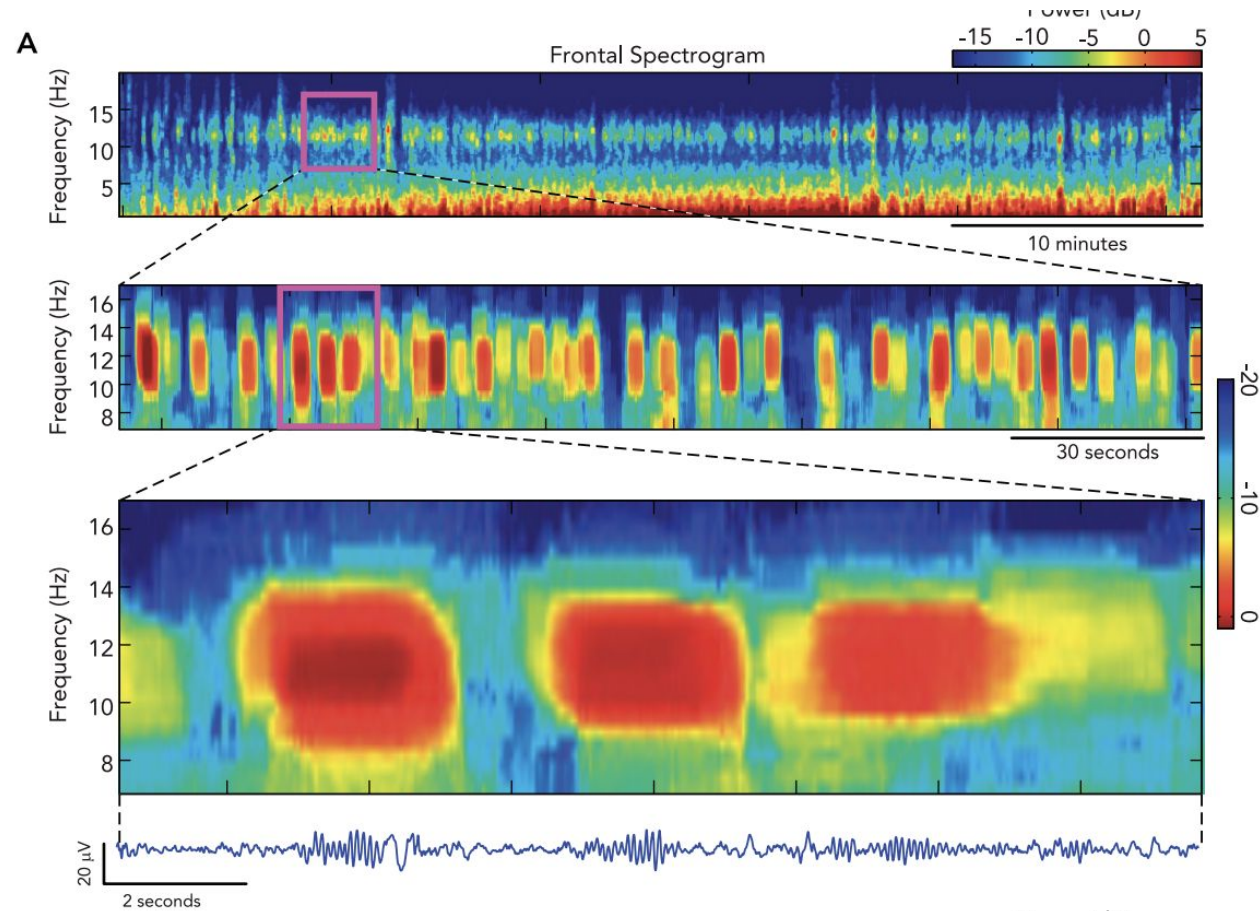


FIGURE 8. The multitaper spectrogram clearly represents spindles as distinct regions of transient spectral power
A multi-scale visualization of the frontal EEG spectrogram shows spindles centered around a single frequency (A) in one subject, and “high” and “low” spindles in another subject (B). By using the multitaper spectrogram, it can be much easier to disambiguate distinct overlapping spindles at different frequencies than in the time-domain traces (bottom).

K-Complex in NREM

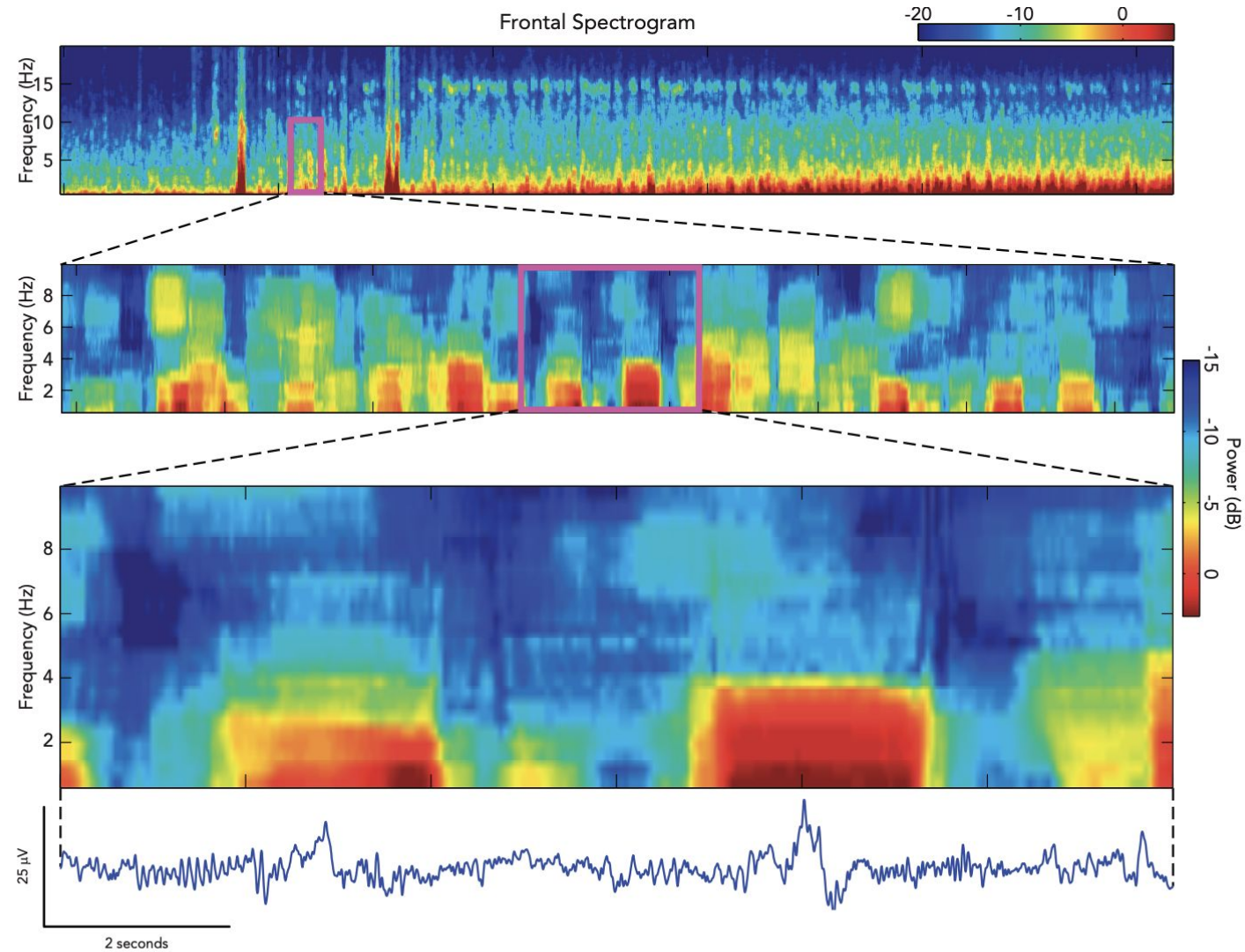
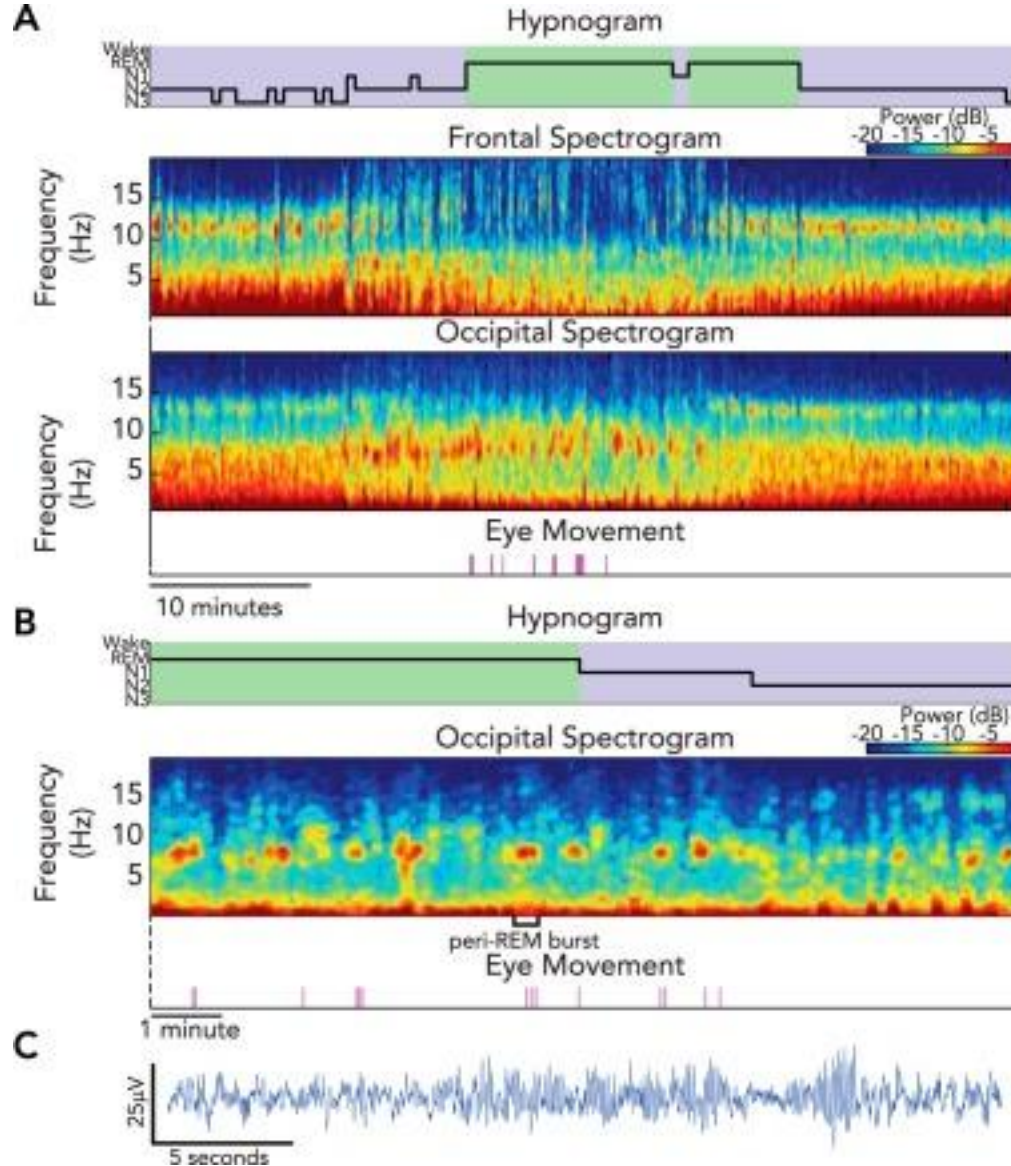


FIGURE 9. The multitaper spectrogram represents K-complex activity as transient low-frequency power. A multi-scale visualization of the spectrogram into the frontal EEG spectrogram shows the spectral signatures of K-complexes during the start of NREM.

REM



THE REM PERIOD

Delta	Rapid decrease to low power following NREM
Theta	Rapid decrease to low power following NREM
Alpha	Recurring transient occipital power in low-alpha frequencies
Sigma	Low power
Background	Background power level above that observed during Eyes Closed Wake

Dynamics

- From NREM, delta and theta power and bandwidth rapidly decrease
- From NREM, sigma power disappears
- Background power is higher than during eyes-closed Wake

Peri-REM Alpha Bursting

- Subjects can exhibit transient, recurring bursts occipitally in low-alpha frequencies
- The bursts:
 - Have lower power, bandwidth, and peak frequency than the alpha seen during eyes-closed Wake
 - Begin to appear at the end of the preceding NREM period and may continue into the next NREM period
 - Decrease in temporal recurrence as REM progresses
 - Burst frequency changes across the peri-REM period

Joining the challenge

- 1) If you don't already have an account, create an account on kaggle.com.
- 2) Join the challenge on:
<https://www.kaggle.com/t/0cddd9911e254661b99e8cdb5049a6b7>
- 3) That's it !
You have access to some notebook to explore data in the data section of the challenge: introduction-to-the-beacon-biosignals-2025.ipynb
- 4) You can have access to the data on your computer by downloading them or directly from kaggle by creating a notebook: <https://www.kaggle.com/docs/notebooks>

Data format

Input data: one .npy file (numpy) per record

Five-channel EEG (C3-M2, F4-M1, F3-F4, F3-M2, and F4-O2) array shaped (5, N_samples) sampled at 100 Hz.

Output data: Integer label vector shaped (N_epochs,).

More details here:

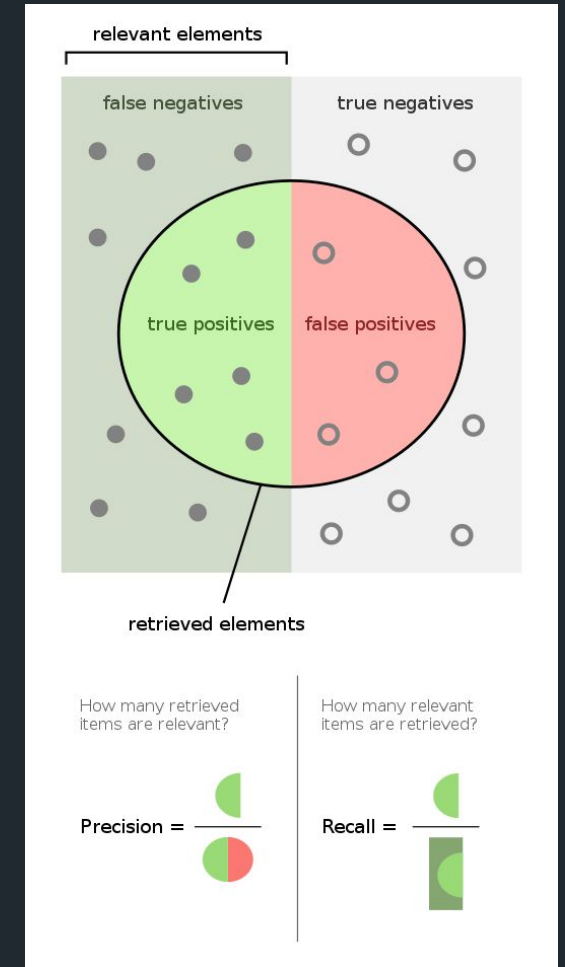
<https://www.kaggle.com/competitions/beacon-biosignals-sleep-staging-2025/data>

Evaluation metric

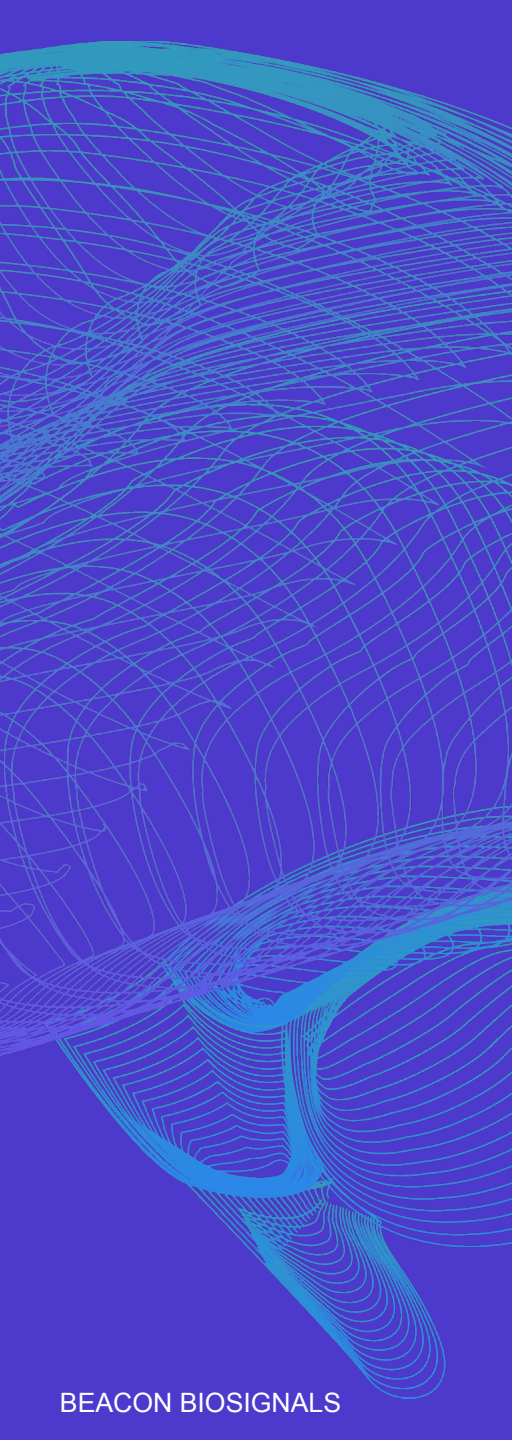
Macro F1-score: Weights equally precision and recall for all classes

$$\text{F1 Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

The MF1 score is the **average of the F1 score** for all classes.



Source:
<https://en.wikipedia.org/wiki/F-score>



Advices

- 1) Have a look at the literature regarding automated sleep staging
- 2) Sleep is a temporal process, you may want to look at the neighbors of each data points and model temporal dependencies.
- 3) Have a look at class balance/imbalance in the dataset
- 4) To open .npy file with R, you can use: <https://github.com/eddelbuettel/rcppcnpy>
- 5) If you need a GPU to train Deep Learning models, you should be able to do so using the accelerator of the Kaggle notebooks.
- 6) If you struggle to do a first submission, rely on the notebook attached in the data

BEACON
BIOSIGNALS

